

UrbanHeat AI

Al Khor area study, Qatar

Theme: Urban Expansion, Land Use Change & Heat Risk

Satellite and local evidence for heat-protection site assessments

Team Al Zubarah, Qatar

UAE Space Academy hackathon, October 2026

The planning problem

Find hot cells, then check who uses them and when.

Satellite screening helps organise visits; local review adds context about shade, access and planned use.

Human exposure has not been measured. Trees, shelters and safer routes are options to assess.

The prototype supports investigation; it does not establish heat-health risk.

Bus-stop site assessment: V04-23

Satellite: 50.4 °C mean of two cell medians; 0% detected green pixels at the current threshold.

Local evidence: roadside trees, no shelter and limited walking access; Nasser confirms site visits.

Next check: waiting-time shade, passenger use, route safety, ownership and feasibility.

300 m values are not stop measurements. Nasser confirms unchanged physical conditions; visit dates unspecified.

Open satellite data and mapped context

Sentinel-2: green-pixel indicators from 5, 15 and 30 September 2026.

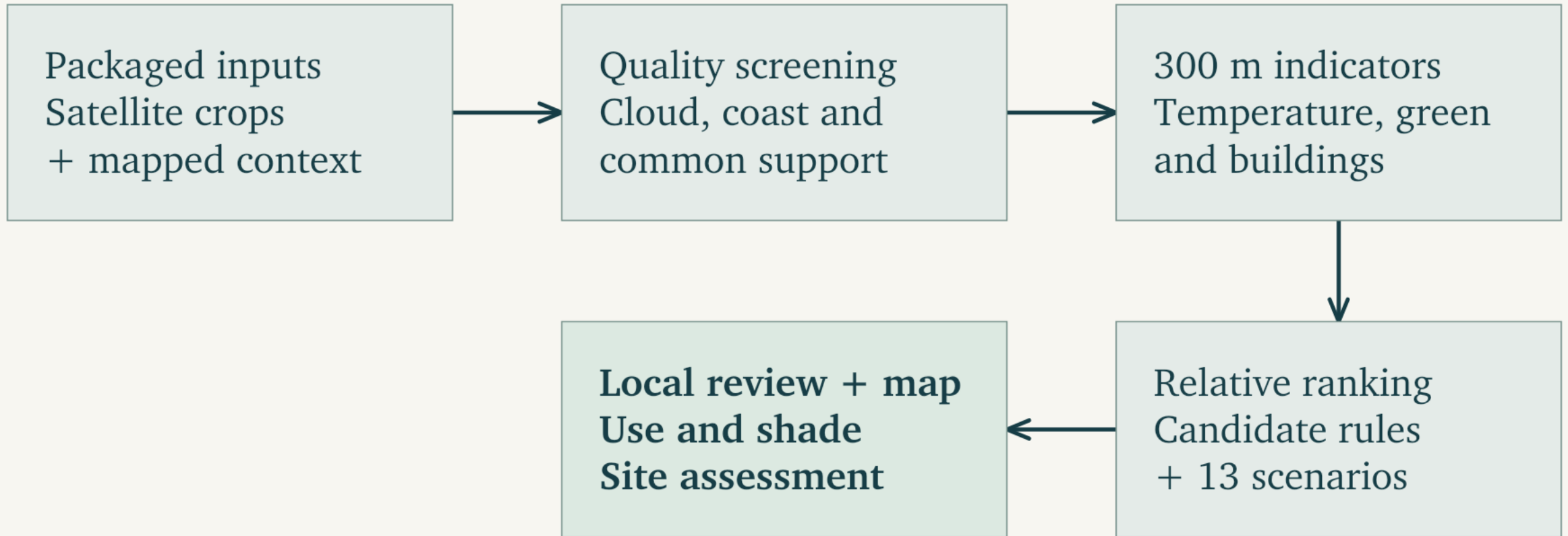
Landsat: surface temperature from 14 and 30 September. The 6 September scene fails coverage QA.

WorldCover 2021: historical land-cover context. OpenStreetMap: contributed building footprints downloaded 5 October.

No hyperspectral, 813 or proprietary VHR data are used.

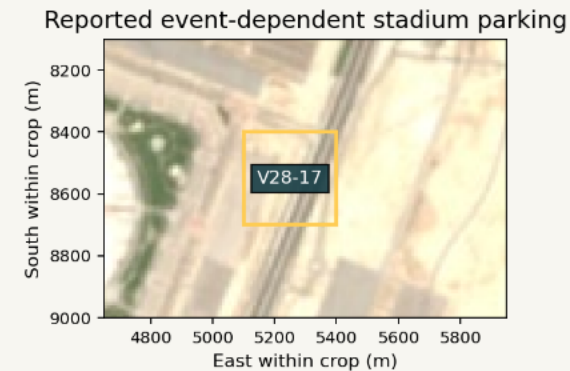
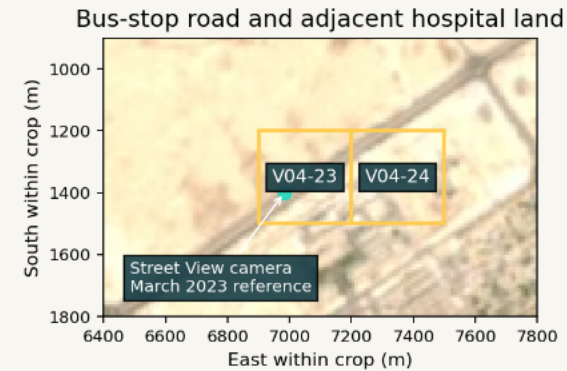
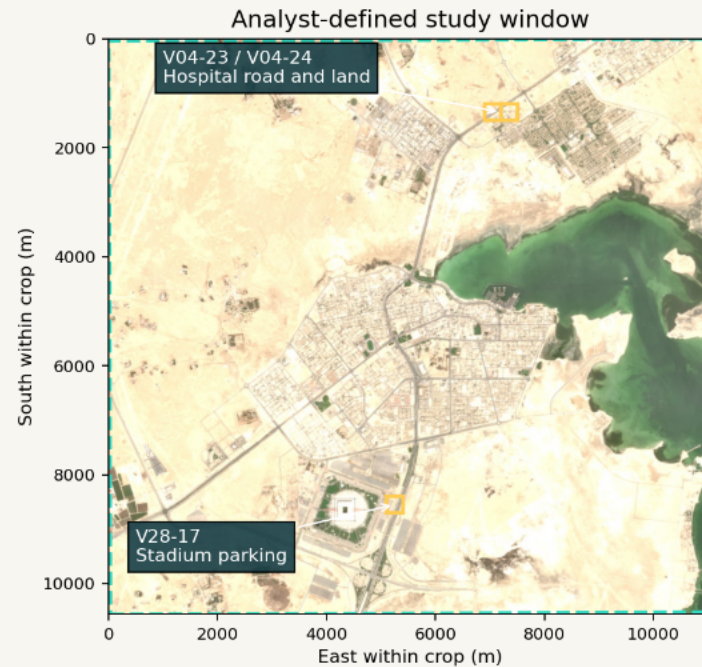
Thermal native detail is about 100 m, delivered on a 30 m grid. We report 300 m cells.

From source crops to an inspectable shortlist



Deterministic processing; upstream WorldCover uses ML. Our own segmentation model is future work.

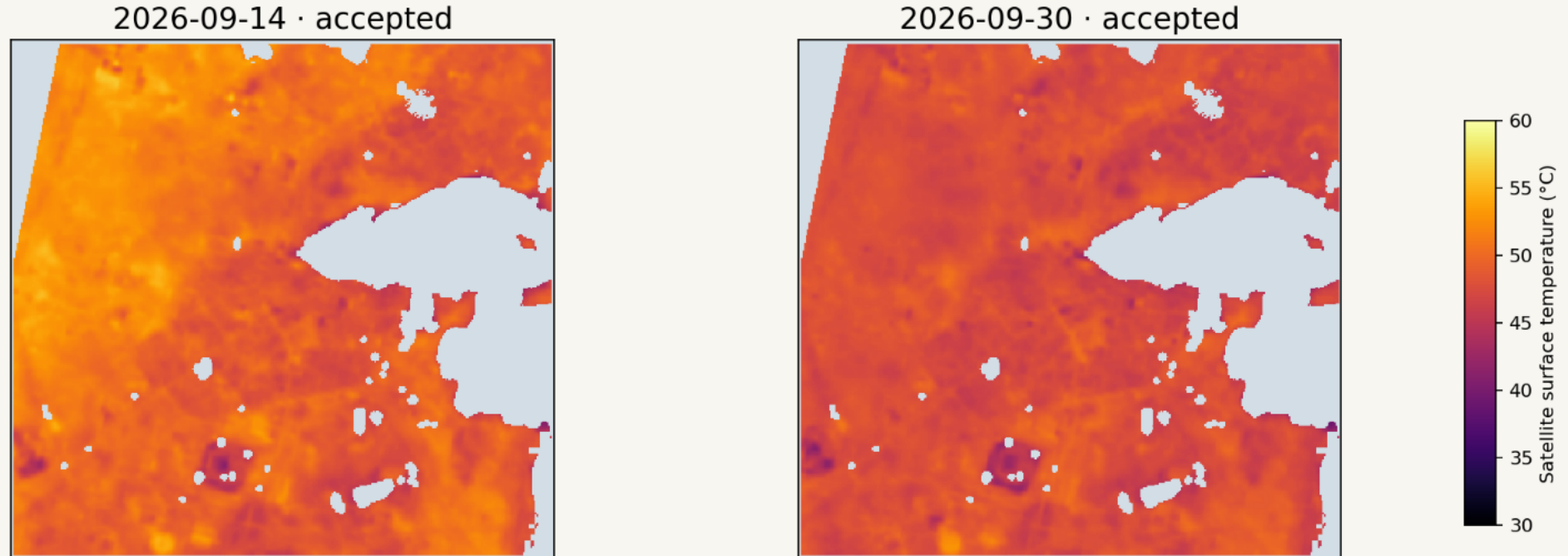
Al Khor study window and shortlist



Base: packaged Sentinel-2, 30 September 2026 · Gold: 300 m cells · Teal dashed: study boundary
Local names and use are team-member observations; visits pending. Camera is not a surveyed stop point. Contains modified Copernicus Sentinel data (2026).

1,111 observed cells and 373 screened urban candidates. Analyst-defined window, approximately 116 km².

Matched surface-temperature observations



6 September excluded: 0% of historical urban inland support passed the strict filters.
94.4379 km² of common inland sample footprints. Grey: excluded or missing observations.

Morning surface observations, not air temperature. Native thermal detail about 100 m, delivered at 30 m.
Date differences do not establish long-term cooling. Landsat imagery courtesy of USGS.

94.4379 km² of common inland sample footprints across the broad study window.

Leading candidates change with the settings

Candidate	Surface °C*	Green %*	Buildings %	Rank range
V04-23	50.4	0.0	2.3	1-14
V28-17	50.0	0.0	0.1	1-23
V04-24	50.2	0.1	2.9	2-34

Bus-stop road: inspect waiting-time shade and access. Parking: check event use.
Hospital land: confirm planned use.

0% detected greenery does not mean no trees. V04-23's NDVI maxima are below 0.30; its ranking stays unchanged.

*Means of date-level cell summaries. Rank ranges are scenario ranges, not confidence intervals.

Reproducibility and reference checks

The package includes 35 hashed input files, source metadata and pinned Python 3.12 dependencies.

Clean GitHub execution passed through a normal Jupyter kernel: five cells, zero errors and all 35 input hashes verified.

Optical rule checks: 24/25 grass points pass; 0/30 building points pass. All 30 coastal points are excluded from thermal support.

Contributed reference labels have conflicts. Human confirmation and independent thermal validation remain pending.

Potential impact and incubation next steps

Completed: reproducible satellite screening, sensitivity checks and local-review examples for site assessment.

Next: detailed shade, use and access checks plus planner feedback; evaluate inspection decisions.

Future: comparable imagery across years, suitable VHR segmentation and verified shade, paving and human-use data.

The current PoC does not measure urban expansion, health risk, causal cooling or long-term warming.

Team responsibilities

Nasser Alhuri: team coordination, site-visit confirmation and submission.

Abdulrahman Almohannadi: supplied local observations for three cells.

Mohammed Almarri: test the map and document issues.

Majed Alkuwari: review slides and explain the limitations.

Ali Alkubaisi: demonstrate the project and prepare judge questions.

Nasser confirms site visits; detailed assessment and other assigned reviews remain pending.

Data attribution and sources

Actual data: Copernicus Sentinel via Earth Search; Landsat via Planetary Computer (USGS); WorldCover 2021; OSM.

Credits: modified Copernicus Sentinel data (2026); Landsat courtesy USGS; WorldCover CC BY 4.0; OSM ODbL.

Discovery and verification resources: Copernicus Browser, NASA Earthdata and Bellingcat toolkit. These are not new analysis sources.

Exact scene provenance and full licences accompany the package. Original code: MIT.